Visual RAG: An Autonomous Visual Search Agent for Video Retrieval

Zhenyu Zhang —Master's Thesis (draft)

Abstract

Video is widely recorded but remains difficult to search at the moment level, especially when a query expresses causal or temporal intent. This dissertation presents Visual RAG, an agentic retrieval-augmented generation system that indexes timestamped visual and transcript evidence, decomposes questions into retrieval-friendly descriptions, searches and re-ranks candidate moments, and answers with grounded timestamp citations. Evaluation on 3,358 temporally grounded questions over 567 NExT-QA/NExT-GQA videos shows that visual retrieval substantially outperforms transcript retrieval. SigLIP with query decomposition nearly doubles corpus-scope R@1 from 0.026 to 0.048. A LangGraph agent improves five-choice accuracy from 0.447 to 0.547, while supplying visual evidence raises temporal-question accuracy from 0.341 to 0.636. The results distinguish recall, ranking precision, and evidence delivery as separate bottlenecks and provide reproducible guidance for video RAG on consumer hardware.

Contents

Chapter 1

Chapter 1 —Introduction

1.1 Motivation

Video has become the dominant medium in which everyday life, education, and work are recorded, yet it remains the least searchable. Text search operates at the granularity of the answer: a query returns the sentence or passage that contains it. Video search, in practice, still operates at the granularity of the *container*: a title, a description, at best a manually chaptered timeline. The question a user actually has —*why did the boy carry the present to the sofa? what did the dog do after it reached the cushion?* —is a question about a particular few seconds of footage, phrased in causal and temporal language that neither keyword matching over metadata nor speech-transcript search can resolve. The gap is felt anywhere video accumulates faster than it can be watched: personal libraries, lecture archives, meeting recordings, broadcast footage, and the industrial video-intelligence platforms that motivated this project's supervision.

Two research threads have recently made this gap addressable. Contrastive vision-language models (CLIP and its successors) embed images and natural language into a shared space, making *what is on screen* directly searchable by description. And retrieval-augmented generation (RAG) shows how a large language model can be grounded in retrieved evidence to answer questions it could not answer alone —with the retrieval unit, in the video case, naturally being a *time-stamped segment* rather than a document. Between these threads sits an under-explored composition: an *agentic* system that decomposes a causal or temporal question, searches a multi-modal index of video segments, reasons over what it retrieves —including reasoning *along the timeline* —and synthesises an answer whose every claim cites the seconds of video that support it.

This dissertation designs, builds, and evaluates such a system end-to-end, under a deliberately modest compute budget (a single consumer laptop GPU plus commodity LLM APIs), and subjects each of its design decisions to quantitative test on a benchmark with ground-truth temporal annotations.

1.2 Research questions

The project plan fixed three research questions, which this dissertation answers in Chapter 5:

- **RQ1** —How effectively can a Visual RAG pipeline retrieve relevant video segments from natural-language queries compared to text-only retrieval baselines?

- **RQ2** —What is the optimal embedding strategy for multi-modal video representation (visual + audio + text)?
- **RQ3** —How can an agentic framework autonomously decompose complex queries into sub-retrieval tasks and synthesise coherent answers?

Temporal precision is treated throughout as a first-class requirement rather than an afterthought: retrieval is scored not on finding the right *video* but on finding the right *moment* (temporal intersection-over-union against ground-truth intervals), and generated answers must carry timestamp citations.

1.3 Contributions

The dissertation makes five contributions, each developed in the chapter noted and empirically grounded in Chapter 5:

1. **An end-to-end, openly reproducible video RAG pipeline with temporal grounding** (Chapter 3): ingestion (keyframes, Whisper speech recognition), overlapping temporal segmentation, dual-modality indexing in a joint vision-language space, staged retrieval, and an answering agent with per-claim timestamp citations —evaluated on 3,358 temporally annotated questions over 567 videos (NExT-QA/NExT-GQA), and runnable end-to-end on one consumer GPU plus approximately US$5 of API usage.

2. **A systematic ablation of the multi-modal embedding strategy** (§5.3), including two documented *negative results* with causal analyses: weighted late fusion of visual and transcript embeddings never outperforms visual-only retrieval at any weight on this domain, and a text cross-encoder re-ranker strictly degrades ranking quality. Both trace to the same cause — sparse, multilingual, conversational speech is a poor retrieval key for everyday video —and both redirect effort toward the visual channel.

3. **A two-stage retrieval equivalence finding** (§5.3.4): re-ranking a cheap index's top-30 with a stronger visual backbone matches indexing the entire corpus with that backbone to within measurement noise, while exchanging the index for SigLIP nearly doubles top-1 moment retrieval —together giving a concrete, measured recipe for scaling quality and cost independently.

4. **An agentic answering layer with temporal tools and bounded self-reflection** (Chapter 3, §5.4): a LangGraph state machine that anchors events by search, walks the timeline around them, and audits its own draft citations against gathered evidence, improving five-way QA accuracy from .447 to .547 over a single-loop agent —and, via a controlled evidence-channel experiment holding all else fixed, a demonstration that supplying keyframes to the answering model nearly doubles accuracy on *what-happened-next* questions (.341 → .636). Modality, the experiment shows, matters independently at both ends of the pipeline.

5. **Reproducible research artefacts**: the complete codebase, evaluation harness, per-experiment result files, provenance log mapping every reported number to the command that produced it, generated figures, and an interactive demonstration interface whose controls map one-to-one onto the ablation dimensions.

1.4 Dissertation structure

Chapter 2 reviews the background: contrastive vision-language pretraining, video moment retrieval and its benchmarks, retrieval-augmented generation, and agentic LLM frameworks. Chapter 3 presents the system design with the rationale for each decision. Chapter 4 records the implementation: the concrete stack and the engineering required to run the full pipeline within the stated budget. Chapter 5 —the core of the dissertation —evaluates the system against the three research questions. Chapter 6 discusses what the results mean, their limitations, and threats to validity. Chapter 7 concludes and outlines future work.

Chapter 2

Chapter 2 —Background and Related Work

This chapter reviews the four research threads the system composes —contrastive vision-language pretraining (§2.1), video moment retrieval and its benchmarks (§2.2), retrieval-augmented generation (§2.3), and agentic LLM frameworks (§2.4) —then surveys the emerging work that, like this project, combines them into LLM-driven video understanding systems (§2.5), and states the gap this dissertation addresses (§2.6). Citation keys refer to `references.bib`.

2.1 Contrastive vision-language pretraining

CLIP (Radford et al., 2021) established that a dual-encoder trained with a contrastive objective on web-scale image–text pairs yields a joint embedding space in which arbitrary natural-language descriptions can be matched against images zero-shot. Two properties of this family underpin the present system. First, the *dual-encoder* architecture separates image and text encoding, so a corpus can be embedded once offline and queried later with only the (cheap) text tower —the architectural premise of Chapter 3's offline/online split. Second, because training pairs are alt-text captions, the encoders are strongest at matching *literal scene descriptions*; abstract, causal, or temporal language falls outside the training distribution, a mismatch this project addresses at query time (§3.4) rather than by fine-tuning.

The open-source replication effort around open_clip demonstrated reproducible scaling laws for this family and released the LAION-trained checkpoints used here (Schuhmann et al., 2022; Cherti et al., 2023). SigLIP (Zhai et al., 2023) replaced CLIP's softmax-based InfoNCE objective with a pairwise sigmoid loss, removing the global normalisation over the batch and yielding markedly stronger retrieval performance at comparable scale. The backbone ablation of §5.3.4 —in which SigLIP SO400M nearly doubles top-1 moment retrieval over CLIP ViT-B —provides an independent, temporally grounded replication of that advantage on everyday video.

2.2 Video moment retrieval, video QA, and temporal grounding

Text-to-video retrieval initially operated at whole-video granularity, with benchmarks such as MSR-VTT (Xu et al., 2016); CLIP4Clip (Luo et al., 2022) showed that frame-level CLIP features, pooled, transfer to this task with minimal video-specific machinery —the same frames-as-proxy assumption this system makes at segment level. *Moment* retrieval sharpened the task to localising an interval inside a video: Gao et al. (2017) introduced the sliding-window formulation and Charades-STA, dense event captions came with ActivityNet Captions (Krishna et al., 2017), and Moment-DETR (Lei et al., 2021) reframed localisation as end-to-end set prediction. The present system inherits the sliding-window formulation —its 8-second overlapping segments are windows scored by embedding similarity —but performs *corpus-level* moment retrieval (find the moment across 567 videos), which moment-retrieval work typically assumes solved by an upstream video search.

On the question-answering side, NExT-QA (Xiao et al., 2021) moved video QA beyond descriptive questions to causal and temporal ones —*why* and *what-happened-next* —over everyday videos. Its extension NExT-GQA (Xiao et al., 2024) added 10.5K temporal grounding labels and reported a sobering finding: state-of-the-art video-language models answer well while attending to the *wrong moments*, i.e. their answers are weakly grounded. This finding motivates both the benchmark choice and the evaluation design of this dissertation: NExT-GQA's interval labels are what allow retrieval to be scored at moment granularity (tIoU) rather than video granularity, and answer grounding —every claim citing its seconds of video —is treated as a first-class output requirement rather than a diagnostic. SeViLA (Yu et al., 2023) addressed grounding by chaining a localiser and an answerer within one trained model; the present system reaches a similar decomposition —localise-then-answer —with zero-shot components orchestrated by an agent instead of end-to-end training.

2.3 Retrieval-augmented generation

RAG couples a parametric language model with a non-parametric retrieval index (Lewis et al., 2020; Guu et al., 2020), classically over dense passage embeddings (Karpukhin et al., 2020). Three RAG sub-problems recur in this dissertation in video form.

Query–document mismatch. When queries and indexed content live in different registers, rewriting the query helps: HyDE (Gao et al., 2023) generates a hypothetical document and searches with its embedding; least-to-most prompting (Zhou et al., 2023) decomposes complex questions into simpler sub-problems. The W5 decomposer (§3.4) transfers this idea across modalities: the LLM rewrites a causal question into the *scene-caption register that a contrastive visual encoder can match*, which is HyDE's manoeuvre with "document" replaced by "caption". Multiple rewrites are combined with reciprocal rank fusion (Cormack et al., 2009), whose rank-based, score-free formulation is what makes fusing differently phrased sub-queries safe.

Precision at the top of the ranking. Two-stage retrieve-then-re-rank pipelines with a cross-encoder second stage are standard in text retrieval (Nogueira and Cho, 2019), with the BGE family (Chen et al., 2024) among the strongest open re-rankers. Chapter 5 shows the transfer to video is *not* automatic: the text cross-encoder degrades ranking on everyday video because transcripts are weak relevance evidence (§5.3.3), and the effective second stage is instead a stronger *visual* encoder —same architecture-level pattern, different modality.

Self-critique. Self-RAG (Asai et al., 2024) trains a model to critique its own retrieval and generation with reflection tokens. The graph agent's reflect node (§3.5) implements the same principle —audit the draft against the retrieved evidence before answering —as a zero-shot prompted pass with an explicit, bounded control-flow position, rather than a trained capability.

2.4 Agentic LLM frameworks

Chain-of-thought prompting (Wei et al., 2022) established that intermediate reasoning improves LLM task performance; ReAct (Yao et al., 2023) interleaved that reasoning with *actions* against external tools, defining the observe – think – act loop that underlies most current agents, including both agents in this project. Toolformer (Schick et al., 2023) demonstrated that tool invocation can be learned self-supervised, foreshadowing the native function-calling interfaces that commercial LLM APIs now expose and on which the W4 agent is built directly. Reflexion (Shinn et al., 2023) added verbal self-feedback across episodes, improving agents without weight updates — the within-episode analogue of which is the reflect node above. This project implements its multi-step agent on LangGraph (a state-machine orchestration library from the LangChain project), chosen because it makes the control flow —nodes, bounded loops, termination conditions —explicit and inspectable (§3.5), in contrast to prompt-encoded loops whose behaviour is implicit in model outputs.

2.5 LLM-driven video understanding and video RAG

A rapidly growing line of work applies LLM orchestration to video. LLoVi (Zhang et al., 2024) shows a deliberately simple recipe —densely caption short clips, then let an LLM aggregate the captions —is strong on long-video QA including NExT-QA. Two contemporaneous systems named VideoAgent make the LLM an active controller: Wang et al. (2024) iteratively *search* for informative frames, answering NExT-QA with ~8 frames on average, while Fan et al. (2024) equip the LLM with a structured memory over events and object tracks, queried through tools. VideoRAG (Ren et al., 2025) scales the RAG framing to extremely long multi-video corpora with a graph-organised index. The present system sits in this family but differs in emphasis in three ways. First, *temporal grounding is the deliverable*: retrieval is evaluated by tIoU against NExT-GQA intervals and answers must cite timestamps, whereas the systems above report QA accuracy alone. Second, it operates at *corpus* scope —the question does not name its video, and finding it is part of the task. Third, its contribution is *measurement*: a component-wise ablation programme (modality, fusion, decomposition, re-ranking, backbone, evidence channel) with documented negative results, rather than a new model or a leaderboard entry.

2.6 Summary of the gap

Each ingredient of this project is established: contrastive encoders for zero-shot visual matching, sliding-window moment retrieval, RAG's retrieve-rewrite-rerank toolkit, ReAct-style tool agents, and LLM video orchestration. What the literature lacks —and what NExT-GQA's grounding finding (Xiao et al., 2024) shows is missing in practice —is an account of how these ingredients compose into a *temporally grounded*, corpus-scope video RAG system, with the contribution of each component measured under one protocol, including where standard components fail to transfer. Providing that account, on reproducible consumer-scale infrastructure, is the role of Chapters 3–6.

Chapter 3

Chapter 3 —System Design

This chapter describes the design of the Visual RAG system and, for each component, the reasoning behind its design choices. The architecture was shaped by one deliberate constraint and one methodological commitment. The constraint is that the online system must run on consumer hardware (a 6 GB laptop GPU): all heavy computation is pushed into a one-time offline indexing stage, and the per-query path touches only a text encoder, an approximate-nearest-neighbour index, and an LLM API. The commitment is that every retrieval component should be *independently measurable*: modalities are indexed separately, pipeline stages are toggleable, and each addition of Chapters 4–5 (decomposition, re-ranking, temporal tools, reflection) can be switched on and off without touching the rest. This is what makes the ablation programme of Chapter 5 possible, and several design choices below are justified primarily by it.

3.1 Architecture overview

Figure 3.1 (fig1_pipeline) shows the system's two lanes. The **offline lane** ingests a video corpus and produces a portable index: keyframes and timestamped transcripts are extracted per video, fused into overlapping temporal segments, embedded into a joint vision-language space, and stored in a vector database. The **online lane** serves a natural-language query: the query is (optionally) decomposed into retrieval-friendly sub-queries, encoded, searched against the index, (optionally) re-ranked by a stronger visual model, and the resulting segments —timestamps, transcripts, and keyframes —are handed as evidence to an answering agent that produces a grounded response with per-claim timestamp citations.

The offline/online split is more than an engineering convenience. It makes the expensive artefacts *reusable and portable* —the index of 567 videos is a few hundred megabytes that can be computed once on any GPU and copied anywhere —and it makes the research loop fast: every experiment in Chapter 5 re-uses the same frozen ingestion outputs, so a full retrieval evaluation over 3,358 questions runs in minutes on a laptop. The split also mirrors production video-search architectures, where indexing throughput and query latency are separate budgets.

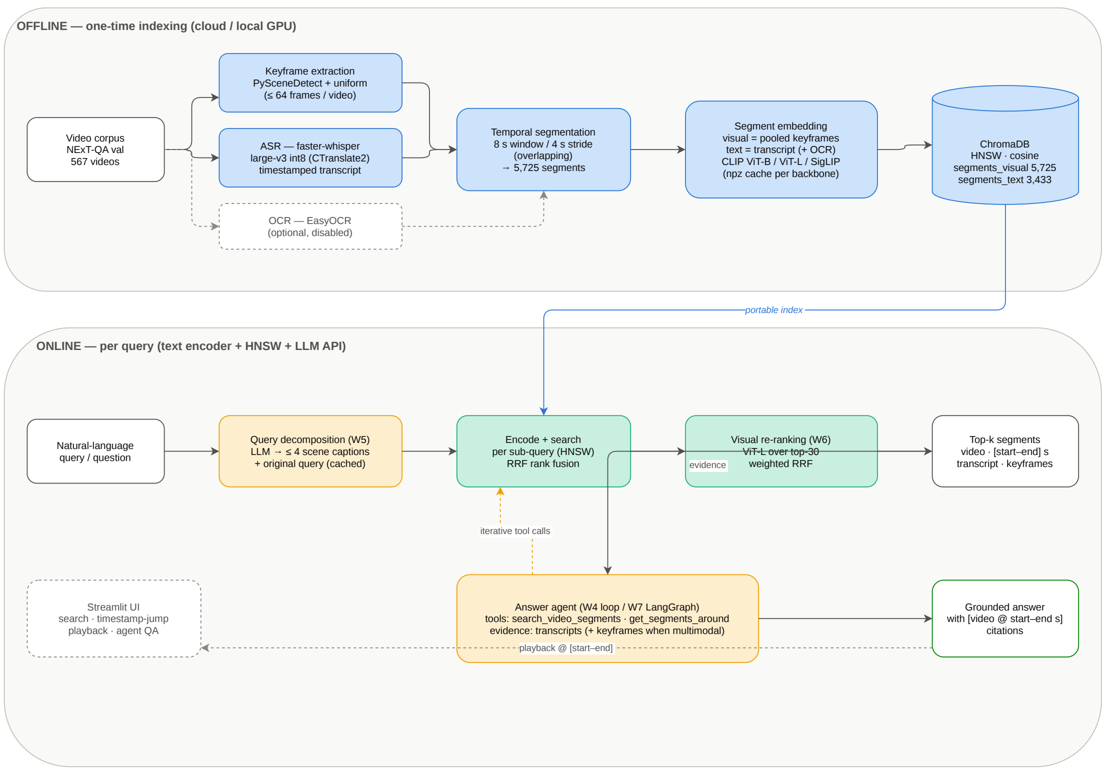

Figure 3.1: Visual RAG offline indexing and online retrieval pipeline.

3.2 Ingestion and temporal segmentation

Keyframes. Each video is decoded once and sampled by two complementary strategies: shot-boundary detection (PySceneDetect's adaptive detector) contributes a representative frame per scene change, and uniform sampling at 0.5 fps guarantees coverage of long static shots that trigger no boundary. Timestamps are de-duplicated with a minimum 1 s gap and capped at 64 frames per video (an even subsample if exceeded); frames are stored as JPEGs resized to 384 px on the long side. The dual strategy reflects the two failure modes of either alone: scene detection under-samples static footage, uniform sampling misses brief transitions.

Speech. Audio is transcribed by Whisper large-v3 in int8 quantisation via CTranslate2, yielding sentence-level chunks with start/end timestamps. Two robustness details matter in practice. First, videos without an audio stream are detected up front and skipped rather than crashing the batch — a real failure mode discovered with silent test clips. Second, language is auto-detected per video and *not* forced to English: NExT-QA's home videos are multilingual, and transcribing them faithfully (rather than mistranscribing them as English) preserves the option of translate-then-embed pipelines later, at the cost of the encoder mismatch analysed in §5.2. On-screen-text OCR is implemented but disabled by default: benchmark footage carries almost no legible text, and the modality can be enabled by configuration where it earns its cost.

Temporal segmentation. The atomic unit of retrieval is the *segment*: an 8-second window advanced with a 4-second stride, carrying every keyframe whose timestamp falls inside it, the concatenated transcript chunks that overlap it, and its exact [start, end] interval. Windows with neither a frame nor text are dropped; the 567-video corpus yields 5,725 segments. Three considerations fix these numbers. The window must be short enough that "retrieving the segment" constitutes a temporally precise answer — 8 s against a mean video length of 44 s gives roughly a five-fold refinement over whole-video retrieval, and is the right order of magnitude for NExT-GQA's ground-truth moments. The 50% overlap ensures that an event near a window boundary appears *whole* in at least one window, at the cost of near-duplicate retrieval results (adjacent overlapping windows frequently co-occur in top ranks, a benign redundancy observed throughout Chapter 5). And fixed windows — as opposed to shot-aligned or content-adaptive ones — keep the ground-truth-to-segment mapping simple and the tIoU ceiling analysable (§6.2). Segment records are persisted as per-video JSONL and are the single interface between ingestion and everything downstream.

3.3 Embedding and dual-modality indexing

Each segment receives up to two vectors in a joint vision-language embedding space. The **visual vector** is the mean of its keyframes' L2-normalised image embeddings, re-normalised — mean-pooling being the standard, hyperparameter-free aggregation that preserves the cosine geometry. The **text vector** embeds the segment's transcript (plus OCR when enabled) with the *same* model's text encoder. Using one joint space for both modalities is what allows a single encoded query to search either collection; the trade-off, accepted knowingly, is that a contrastive text encoder trained on alt-text is a weak reader of long or non-English transcripts (its 77-token context truncates long passages), which Chapter 5 quantifies.

The encoder itself sits behind a thin interface (open_clip) so that the backbone is a configuration value: ViT-B-32 for development, ViT-L-14 and SigLIP SO400M for the Chapter 5 ablations. Every backbone's segment embeddings are cached as per-video arrays on disk, making a backbone swap a one-time re-embedding job and an index rebuild a matter of minutes —the property that made the W8 backbone comparison affordable.

Vectors live in ChromaDB —an embedded vector store chosen over server-class alternatives (Milvus, Qdrant) because at 10^3–10,000 segments HNSW search is effectively exact and operational simplicity dominates. The two modalities are stored as **separate collections** sharing segment IDs and metadata (`video_id`, `start`, `end`, frame count) rather than as one concatenated vector. This is the single most consequential indexing decision: separate collections mean each modality can be searched, scored, and evaluated in isolation, which is precisely what RQ2's ablations require; late fusion, early fusion, or either modality alone all become query-time choices over the same index. Segment text rides along as the collection's document payload, so downstream components (re-rankers, agents) can access it without re-reading disk.

3.4 Retrieval

The retriever exposes one code path used identically by the evaluation harness, the agent, and the demo UI —deliberately, so measured numbers describe the system users interact with.

Single-query search encodes the query with the index backbone's text encoder and performs cosine search in one modality, or in both with **late fusion** (score = $\alpha \cdot$ visual + $(1-\alpha) \cdot$ text, candidates over-fetched threefold before fusion). Late fusion was preferred over early (concatenation) fusion because it requires no joint training, is trivially ablatable per modality, and exposes α as an interpretable knob —the knob the α-sweep of §5.3.1 then showed to have no useful setting on this corpus, itself a finding the design made cheap to obtain.

Query decomposition (W5) addresses the mismatch between causal/temporal question phrasing and CLIP's training distribution of literal scene captions. An LLM rewrites the question into at most four short present-tense captions; each caption *plus the original question* is searched independently, and the rankings are combined by reciprocal rank fusion ($k = 60$). Two design details are load-bearing. Retaining the original question guarantees decomposition can only add candidates, never replace the baseline behaviour —a safety property that made it deployable by default. And rank-based fusion (rather than score averaging) sidesteps the incomparability of cosine scores across differently phrased sub-queries. Decompositions are cached on disk keyed by question text, so the marginal cost across the 3,358-question evaluation was a one-off ~US$0.15.

Second-stage re-ranking (W6) re-scores a 30-candidate pool and re-orders it by a weighted RRF of the first-stage ranking and the re-ranker's ranking. The design went through one documented failure: the planned text cross-encoder (question × transcript) *degraded* every metric, for reasons diagnosed in §5.3.3, and survives only as an ablation configuration. Its replacement re-scores candidates *visually*, with a stronger backbone (ViT-L-14) over pre-computed segment embeddings —turning the re-ranker into a second opinion from a better witness rather than a different modality. One methodological lesson from the failure is baked into both re-rankers: when a candidate lacks the re-ranker's modality (no transcript; no cached vector), its re-ranker rank is *imputed from its retrieval rank* instead of being treated as absent, keeping all candidates on one score scale. Without this, every candidate the re-ranker could score would structurally outrank every candidate it could not —which in an early implementation silently drove R@1 to zero.

3.5 Agentic answering

The answering layer converts retrieved segments into a natural-language answer with per-claim citations of the form `[video_id @ start–end s]`. It is built as native LLM function-calling over two tools, in two harnesses of increasing sophistication.

Tools. `search_video_segments(query, modality)` wraps the retriever; its description instructs the model to phrase queries as scene descriptions rather than questions —pushing the decomposition insight into the agent's tool-use behaviour. `get_segments_around(video_id, timestamp, direction, n)` is the temporal primitive: it returns the segments immediately before or after a timestamp in one video. Temporal questions ("what did X do *after* Y?") are answered by composition —anchor Y with search, then walk `after` —a two-step plan the system prompt states explicitly, and which the agents were observed to follow (§5.4.1).

Simple agent (W4). A bounded tool-use loop: the model may call tools for up to three rounds, after which a final call with tool choice disabled forces an answer from the evidence gathered. The forced-answer step exists because both LLM providers, left to themselves, will occasionally keep requesting searches until the round budget expires and return an empty answer —an early bug whose fix (an explicit "budget exhausted, answer now"turn) applies to both harnesses.

Graph agent (W7). The LangGraph state machine of Figure 3.2 (fig2_langgraph) adds two things a linear loop cannot express cleanly. First, an explicit **reflect node**: when the agent produces a draft answer, a separate LLM pass audits it against the *actually gathered* evidence —do the cited segments exist in the tool results? are the claims supported rather than invented? is the temporal relation to the anchor correct? —and either accepts it (terminating the graph) or returns it once with a critique appended, resuming the tool loop. Reflection is bounded (one revision) and budget-aware: a revision is only requested if enough tool rounds remain to act on the critique, a guard added after observing budget-exhausted revision loops. Second, the graph makes the control flow inspectable: every run yields a typed trace of rounds, tool calls, and reflections, which Chapter 5 uses for its qualitative analysis.

Provider abstraction and the evidence channel. Both agents run against either an OpenAI-compatible endpoint (DeepSeek; any local server such as Ollama) or the Anthropic API. The abstraction is not merely operational: the two providers differ in the *evidence channel*. Under the multimodal provider, tool results embed each segment's middle keyframe as an image block alongside its transcript, capped at six images per result; under text-only providers the same renderer emits transcript and timestamps only. Because the tools, prompts, and retrieval are identical across providers, switching the provider is a controlled manipulation of what evidence the answerer can perceive —the design property that enables the evidence-channel experiment of §5.4.2 to be run as a true single-variable comparison.

3.6 Demonstration interface

A Streamlit application exposes the full pipeline interactively in two tabs. The *search* tab runs the retrieval stack with per-component toggles —modality, query decomposition, visual re-ranking — and renders each hit with its keyframe, timestamps, transcript, and score; expanding a hit plays the source video seeked to the segment's start time, closing the loop from natural-language query to the exact moment on screen. The *ask* tab runs either agent under either provider and displays the answer alongside its full evidence trail (every tool call and the segments it returned). The toggles

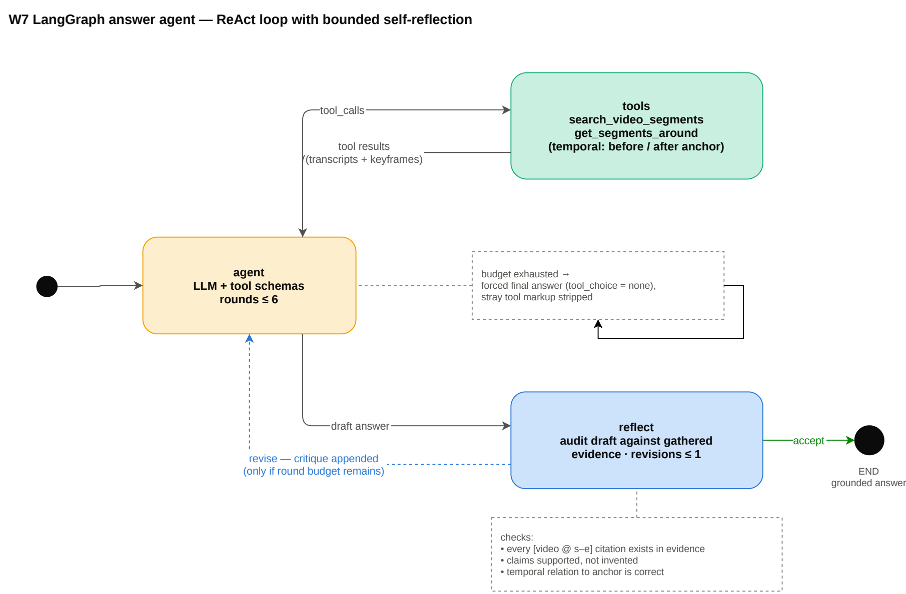

Figure 3.2: LangGraph answering-agent state machine with tools and bounded reflection.

map one-to-one onto the ablation dimensions of Chapter 5; the interface therefore doubles as a research instrument —the qualitative observations cited in Chapters 5 and 6 were made through it —and as the project's demonstrable artefact.

3.7 Summary

The design commits to cheap, measurable components over end-to-end training: fixed overlapping segments as the unit of temporal grounding; one joint embedding space with modalities indexed separately; retrieval improvements (decomposition, re-ranking) that compose as pure query-time re-rankings; an agent layer whose tools, control flow, and evidence channel are each independently switchable. Chapter 4 details the implementation of these components; Chapter 5 measures them.

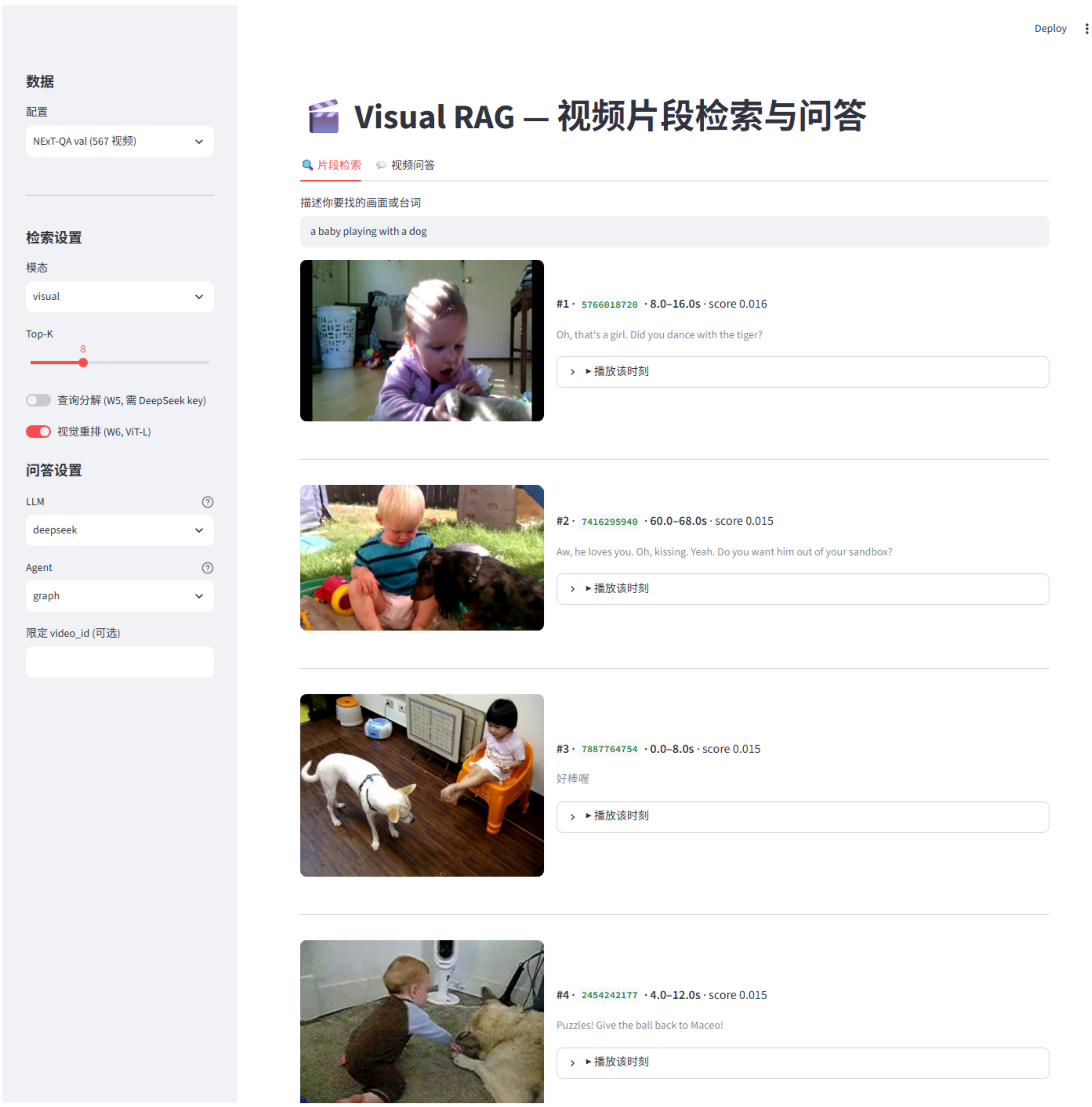

Figure 3.3: Visual RAG demonstration interface showing search results and timestamped evidence.

Chapter 4

Chapter 4 —Implementation

Chapter 3 described *what* was built and why; this chapter records *how*, with emphasis on the engineering that made a full benchmark evaluation feasible on a single consumer laptop (RTX 4050, 6 GB VRAM) and a few dollars of API usage. The codebase is a small Python package (`visualrag/`) with thin CLI entry points (`scripts/`), one YAML configuration per experimental condition, and a Streamlit interface (`ui/`).

4.1 Technology stack

Concern	Choice	Notes
Video decoding / duration	PyAV	single decode pass per video
Shot detection	PySceneDetect (adaptive)	+ 0.5 fps uniform fallback
Speech recognition	faster-whisper (large-v3, int8)	CTranslate2 runtime
Embeddings	open_clip: ViT-B-32, ViT-L-14, SigLIP SO400M	one interface, three backbones
Vector store	ChromaDB (embedded, HNSW, cosine)	two collections per backbone
Re-ranker (ablation)	sentence-transformers bge-reranker-v2-m3	text cross-encoder, §5.3.3
Agent framework	LangGraph	W7 state machine
LLM providers	DeepSeek, Anthropic Claude, any OpenAI-compatible endpoint	provider-agnostic answerer
UI	Streamlit	search + timestamp-jump playback
Evaluation	custom harness (`visualrag/eval`)	all metrics of §5.1

Two stack decisions deserve one sentence each. ChromaDB was chosen over server-class vector databases because at $10^3 - 10{,}000$ vectors, HNSW is effectively exact and zero operational overhead beats theoretical scalability. LangGraph was chosen over a heavier agent framework because the W7 design needed exactly one thing —an explicit, bounded, inspectable state machine —and nothing else.

4.2 Fitting the pipeline into 6 GB

The plan's premise (§3.1) was that heavy computation happens offline; the implementation's contribution was making "offline"affordable *locally* rather than requiring a cloud GPU. Three techniques mattered.

Decoupling ASR from the deep-learning runtime. Whisper large-v3 nominally exceeds a 6 GB budget under PyTorch, but faster-whisper executes it through CTranslate2 with int8 weights in approximately 3 GB. One integration subtlety cost a bug fix: the pipeline's device auto-detection originally probed *PyTorch's* CUDA availability, which is false under a CPU-only PyTorch build even when the GPU is usable —so the transcriber now probes CTranslate2's own CUDA device count, and ships the pip-installed cuBLAS/cuDNN libraries to the DLL search path on Windows. With this in place the full 567-video corpus transcribed in approximately 2 h locally.

Embedding caches as the unit of reuse. Every backbone's segment embeddings persist as per-video compressed arrays (`.npz`) keyed by segment ID. Indexing reads only the cache; evaluation reads only the index. Re-running any experiment therefore costs seconds; adding a backbone costs one embedding pass; and the visual re-ranker (§3.4) is *free at query time* beyond a matrix product, because its ViT-L vectors are the same cache the ViT-L index ablation used. A related robustness detail: SigLIP's open_clip implementation exposes no `visual.output_dim`, so the encoder wrapper resolves the embedding width by a three-step probe (attribute → model field → dummy forward pass) —the kind of small fix on which "one interface, three backbones"actually depends.

Idempotence everywhere. Every batch stage —ingestion, embedding, indexing, decomposition —checks for its own output and skips work already done, at per-video or per-question granularity. Long jobs on a laptop *will* be interrupted (sleep, OOM, session resets); with idempotent stages, every restart resumes where it stopped. During the project this was exercised repeatedly: the full-corpus ingestion survived two process kills and completed under a self-healing watchdog script that relaunches the batch until its completion marker appears (`scripts/ingest_watchdog.ps1`).

4.3 Cost engineering for LLM-dependent components

Two pipeline components call LLM APIs; both were engineered so that *evaluation scale* never multiplies *API cost*.

Query decomposition results are cached on disk keyed by the question string. The full validation split (3,358 questions, 16 concurrent workers) cost approximately US$0.15 *once*; every subsequent experiment —the W5/W6/W8 ablations each rerun decomposition-dependent retrieval several times —replayed the cache for free. The cache is committed to the repository, making the retrieval experiments deterministic and reproducible without any API key.

Agent evaluation batches questions across a thread pool (the retrieval stack is read-only and thread-safe), parses the model's final option choice from a fixed sentinel line, and records per-question token usage. The 150-question agent comparison cost approximately US$0.35

(DeepSeek); the 44-question multimodal run approximately US$3 (Claude). Rate limits are respected by capping concurrency rather than by retry storms.

4.4 Provider-agnostic answering and its pitfalls

The answerer runs unchanged against Anthropic's API, DeepSeek, or any OpenAI-compatible endpoint (including a local Ollama server —the hook for the open-source-model comparison left as future work). The abstraction is a thin dispatch: one message-format adapter per provider family, one shared tool schema, one shared evidence renderer that adds keyframe image blocks when the provider supports vision. Field experience contributed three fixes worth recording, since each silently produces *empty or corrupt answers* rather than errors. (i) Models sometimes spend the entire round budget requesting more searches; the loop must end with a forced-answer turn with tool use disabled. (ii) Under a forced-answer turn, DeepSeek occasionally emits its intended tool calls as literal markup in the message text; the answer extractor strips these. (iii) When self-reflection can send the agent back for more evidence, the revision path must check the remaining round budget, or the reflect/act cycle consumes it and the graph terminates answerless.

4.5 Evaluation harness and provenance

The harness (`scripts/evaluate.py`, `scripts/eval_qa.py`) exposes every experimental axis of Chapter 5 as a flag (`--scope`, `--modalities`, `--tau`, `--decompose`, `--rerank`, `--agent`, `--provider`) over per-condition YAML configurations, and emits a self-describing JSON per run. All result files are committed under `results/`; `docs/EXPERIMENTS.md` maps every number reported in this dissertation to the exact command, configuration, and date that produced it; and the dissertation's data figures are generated from those JSONs by a committed script (`docs/figures/make_figures.py`), so no reported value exists only in prose. Figures 3.1 – 3.2 (architecture) are maintained as editable draw.io sources alongside their exports.

4.6 Summary

The implementation's through-line is *reuse*: one decode pass per video, one embedding pass per backbone, one LLM call per unique question, one code path shared by harness, agent, and UI. That discipline —more than any individual optimisation —is what let a dissertation-scale evaluation programme (twenty-odd full-split retrieval runs, three agent studies, three backbones) execute on one laptop within a single project week.

Chapter 5

Chapter 5 —Evaluation

This chapter evaluates the system against the three research questions. Section 5.1 describes the experimental setup shared by all experiments. Section 5.2 addresses RQ1 by comparing multi-modal retrieval against a text-only baseline. Section 5.3 addresses RQ2 through a series of ablations over the embedding and retrieval strategy —modality fusion, query decomposition, second-stage re-ranking, and the embedding backbone —including two negative results that shaped the final design. Section 5.4 addresses RQ3 by measuring end-to-end question-answering accuracy of the agentic pipeline, culminating in a controlled experiment that isolates the contribution of visual evidence at the answering stage. Section 5.5 reports system-level costs. Every number in this chapter is reproducible from the committed result files (`results/*.json`); the exact command that produced each result is recorded in `docs/EXPERIMENTS.md`.

5.1 Experimental setup

Dataset. All experiments use the validation split of NExT-QA (Xiao et al., 2021), a benchmark of causal and temporal multiple-choice questions over short (~44 s) everyday videos, together with the temporal grounding labels contributed by NExT-GQA (Xiao et al., 2024). The split contains 3,358 questions over 567 videos, and —after intersecting with the grounding annotations —every question carries at least one ground-truth temporal interval locating the evidence for its answer. NExT-GQA was chosen over larger retrieval benchmarks (e.g. MSRVTT, ActivityNet Captions) precisely because these interval labels allow retrieval to be scored with *temporal* precision rather than at whole-video granularity.

Indexing. The ingestion pipeline of Chapter 3 produced 5,725 overlapping segments (8 s window, 4 s stride) from the 567 videos. Each segment is indexed in two ChromaDB collections (HNSW, cosine): a visual collection holding the pooled CLIP embedding of the segment's keyframes (5,725 vectors), and a text collection holding the CLIP text embedding of its speech transcript (3,433 vectors —60% of segments contain speech). Unless stated otherwise the index backbone is CLIP ViT-B-32 (laion2b); Section 5.3.4 varies the backbone.

Retrieval metrics. For each question, the question text is used as the query and the top-K segments are retrieved. A retrieved segment counts as a *hit* if it comes from the ground-truth video **and** its temporal interval overlaps a ground-truth interval with tIoU $\geq \tau$; requiring both conditions ties the metric to moment-level retrieval rather than video identification. We report Recall@{1,5,10} and MRR at the primary threshold $\tau = 0.5$, nDCG@10 with graded tIoU gains (ideal-normalised against the ground-truth video's own segments), and tIoU@1 —the mean temporal overlap of the top-ranked segment, the dissertation's "temporal accuracy" measure. Section 5.3.5 repeats key configurations at the looser $\tau = 0.3$.

Scopes. Two retrieval scopes are evaluated. In **corpus** scope the query searches all 5,725 segments across 567 videos —the full RQ1 setting, where the system must find both the right video and the right moment. In **video** scope the search is restricted to the ground-truth video's own segments, isolating pure temporal localisation. The gap between the two scopes attributes error to cross-video confusion versus within-video imprecision.

QA metric. Section 5.4 evaluates answer generation directly on NExT-QA's five-way multiple-choice format: the agent receives the question and its five options, gathers evidence with its tools, and must emit a final option index. Accuracy is reported overall and per question type (CW/CH: causal *why*/*how*; TN/TC/TP: temporal *next*/*current*/*previous*). The random-guess floor is 0.20.

Implementation constants. Unless varied by the ablation, fusion weight $\alpha = 0.5$, retrieval depth $K = 10$, re-ranker candidate pool 30, and the question encoder matches the index backbone. All retrieval experiments are deterministic given the committed decomposition cache; agent experiments use temperature-default API calls and are therefore subject to normal LLM variance.

5.2 RQ1 —Multi-modal retrieval against a text-only baseline

Table 5.1 reports the three single-strategy configurations on both scopes at $\tau = 0.5$. The text-only baseline embeds each segment's Whisper transcript and searches it with the question text —the standard "search what was said" approach that RQ1 asks us to beat.

Table 5.1 —Retrieval by modality (ViT-B-32 index, $\tau = 0.5$).

Scope	Modality	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
corpus	visual	.026	.075	.111	.047	.117	.035
corpus	text	.002	.005	.011	.004	.011	.003
corpus	fused (α=.5)	.002	.010	.013	.005	.018	.007
video	visual	.148	.394	.491	.250	.659	.203
video	text	.080	.232	.305	.143	.390	.117
video	fused (α=.5)	.129	.384	.491	.235	.639	.186

Three observations answer RQ1. First, **visual retrieval dominates text retrieval by an order of magnitude in corpus scope** (R@10 .111 vs .011): to find a moment among 567 videos of everyday footage, what is *on screen* is far more discriminative than what is *said*. Second, the failure of the text channel has an identifiable cause rather than being an artefact of poor transcription: NExT-QA's home videos contain sparse, conversational, and frequently non-English speech (German, Mandarin and other languages appear in the transcripts), while the CLIP text encoder that embeds both queries and transcripts is trained predominantly on English alt-text. The information is often simply absent —a question about what a child *does* is rarely answered by what anyone *says*. Third, the corpus/video gap localises the difficulty: within the correct video the system already ranks a correct moment in its top five for 39% of questions, but across the corpus R@10 falls to 11%

—**cross-video confusion, not within-video imprecision, is the dominant error mode**, which motivates the precision-oriented interventions of Section 5.3.

The by-type breakdown (Appendix B) shows temporal-precise question types (TP, n = 52) retrieving worst in corpus scope, previewing the temporal-reasoning theme of Section 5.4.

5.3 RQ2 —Embedding and retrieval strategy ablations

5.3.1 Negative result: naive late fusion never beats visual-only

The natural response to a weak-but-nonzero text channel is weighted late fusion: score = α · visual + (1−α) · text. Figure 5.1 (fig4_alpha_sweep) sweeps α over {0.5, 0.7, 0.8, 0.9} in corpus scope. Every metric rises monotonically toward the α = 1.0 endpoint —visual-only —and none crosses it; the same holds in video scope, where the best fused configuration (α = 0.8) ties visual-only on R@10 (.494 vs .491) but loses on R@1 (.139 vs .148) and MRR (.244 vs .250). At α = 0.5 fusion is catastrophic in corpus scope (R@10 .013) because the text scores act as high-variance noise added to a weak-but-real visual signal.

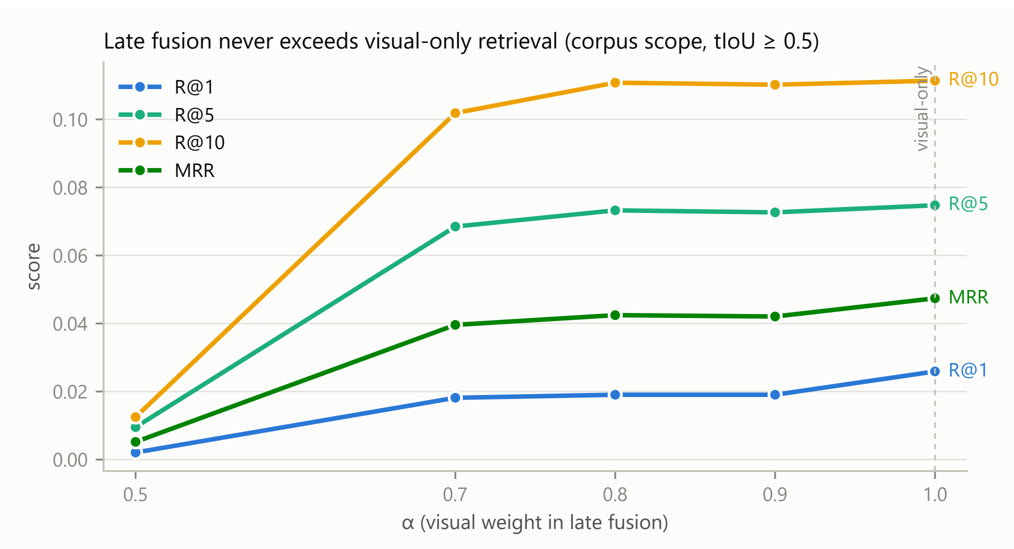

Figure 5.1: Late-fusion weight sweep in corpus scope.

The conclusion is sharper than "tune α": **with this text channel, there is no fusion weight at which the transcript embeddings contribute positive evidence.** Fusion is not a weighting problem but a channel-quality problem —the actionable implication being to repair the text channel (e.g. translate transcripts to English before embedding, or use a dedicated multilingual text-retrieval encoder) before re-introducing it, rather than to search the weight space further. All subsequent experiments therefore use the visual modality.

5.3.2 Query decomposition recalls more, but no more precisely

NExT-QA questions are causal and temporal ("why did the boy pick up one present from the group of them and move to the sofa"), whereas CLIP matches literal scene descriptions. The W5 decomposer prompts an LLM to rewrite each question into at most four short scene captions ("a boy picks up a present from a group of presents”, “a boy carries a present and walks to a sofa”, ⋯), retrieves each caption plus the original question independently, and fuses the rankings with reciprocal rank fusion. The rewrite is computed once per question and cached (3,358 questions approximately US$0.15 of LLM usage).

Decomposition lifts the deep-rank metrics in corpus scope —R@10 .111 → .122 (+9.9% relative), R@5 +6.7%, MRR +6.4% —while leaving R@1 and tIoU@1 unchanged (Table 5.2, rows 1−2; Figure 5.2). The pattern is informative: translating abstract questions into CLIP-native captions surfaces *additional correct candidates* into the top ten, but rank fusion of near-tied cosine scores cannot push them to position one. Recall problems and precision problems are separable; decomposition solves a recall problem, and the next section supplies the missing precision.

5.3.3 Second-stage re-ranking: a negative and a positive result

Text cross-encoder (negative). The project plan anticipated a standard text cross-encoder re-ranker (bge-reranker) scoring (question, transcript) pairs over the candidate pool. On a 100-question development subset this *reduced* quality at every fusion weight tested —at weight 0.5 it halved R@5 (.070 → .030), and even at 0.8 it remained below the no-reranking baseline. The mechanism mirrors §5.3.1: the transcripts are conversational chatter, largely uncorrelated with the visual events the questions ask about, so every reordering the cross-encoder makes injects noise. An implementation subtlety compounds the problem and is worth recording: 40% of segments have no transcript at all, and if such candidates simply receive no re-ranker score, any scored candidate the cross-encoder likes outranks *every* unscored one —structurally demoting exactly the visual-only segments most likely to be correct. (Our first implementation had this flaw and drove R@1 to zero.) The repaired design imputes the re-ranker term of a modality-missing candidate from its retrieval rank, keeping both candidate classes on one score scale; the negative result above is reported *with* this repair, so it reflects the channel, not the bug.

Visual re-ranker (positive). The replacement re-ranker re-scores the top-30 candidates with a stronger vision model (CLIP ViT-L-14) over pre-computed segment embeddings, fusing the ViT-L ranking with the first-stage ranking by weighted RRF. Every segment has keyframes, so the missing-modality problem vanishes. Table 5.2 and Figure 5.2 (fig5_ablation_ladder) show the ladder at $\tau = 0.5$, corpus scope:

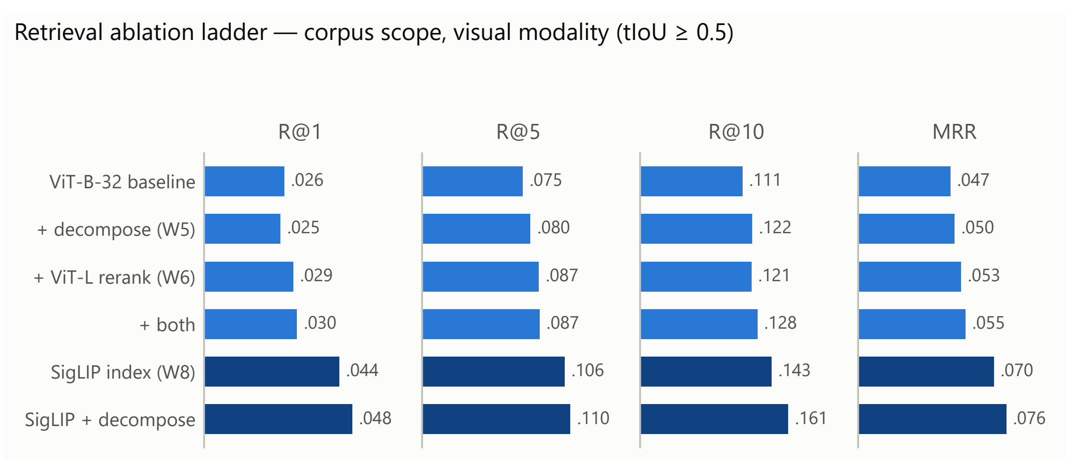

Figure 5.2: Retrieval ablation ladder from the baseline to the final configuration.

Table 5.2 —Corpus-scope ablation ladder (visual modality, τ = 0.5).

Configuration	R@1	R@5	R@10	MRR	tIoU@1
ViT-B-32 baseline	.026	.075	.111	.047	.035
+ decomposition (W5)	.025	.080	.122	.050	.035
+ ViT-L re-ranking (W6)	.029	.087	.121	.053	.041
+ both	.030	.087	.128	.055	.040
SigLIP index (W8)	.044	.106	.143	.070	.054
SigLIP + decomposition	**.048**	**.110**	**.161**	**.076**	.055

The two mid-pipeline interventions compose additively: decomposition contributes at depth, re-ranking at the top of the list, and their combination improves every metric over the baseline (+15–17% relative on MRR and R@10). In video scope the combined configuration reaches R@1 .154, R@10 .501, tIoU@1 .209 —each the best of any ViT-B configuration.

5.3.4 Backbone: SigLIP dominates, and two-stage retrieval matches one-stage

The final RQ2 axis exchanges the embedding backbone itself, holding segmentation and evaluation fixed. Three backbones were compared as *the index*: CLIP ViT-B-32 (baseline), CLIP ViT-L-14, and SigLIP SO400M (webli), each with its own question encoder.

Two findings emerge from Table 5.2 (bottom rows) and Table 5.3. First, **SigLIP is categorically stronger for this task**: as the index it lifts corpus R@1 by 69% relative over ViT-B (.026 → .044) and tIoU@1 by 54%, and with decomposition on top reaches R@1 .048 —**nearly double the original baseline** —and video-scope tIoU@1 .230. This is consistent with SigLIP's sigmoid-loss pretraining, which has been reported to favour retrieval tasks (Zhai et al., 2023).

Second, ViT-L used as the full index (corpus R@1 .029, R@5 .090, R@10 .122, MRR .054) is statistically indistinguishable from ViT-B *plus* ViT-L re-ranking (.029/.087/.121/.053) —every metric agrees within ±.003. **The two-stage cheap-index/expensive-re-ranker design recovers essentially all of the larger model's quality** while embedding the corpus only with the small model; at larger corpus scales, where re-embedding everything with the expensive model is prohibitive, the two-stage design is the scalable route to the same accuracy. A corollary worth stating for practitioners: a re-ranker must *differ* from the index model —re-scoring with the index's own embeddings is an identity operation.

Table 5.3 —Video-scope backbone comparison (visual modality, τ = 0.5).

Index backbone	R@1	R@5	R@10	MRR	tIoU@1
ViT-B-32	.148	.394	.491	.250	.203

Index backbone	R@1	R@5	R@10	MRR	tIoU@1
ViT-L-14	.151	.407	.496	.257	.210
SigLIP SO400M	.170	.406	.494	.266	.221
SigLIP + decomposition	**.176**	**.419**	**.496**	**.275**	**.230**

5.3.5 Threshold sensitivity

$\tau = 0.5$ is a strict criterion for 8-second windows scored against ground-truth intervals of arbitrary length. Table 5.6 repeats the ViT-B baseline and the best ViT-B configuration at $\tau = 0.3$; the binary metrics roughly double, while tIoU@1, which is threshold-free, is unchanged by construction.

Table 5.6 —Threshold sensitivity (visual modality, ViT-B index).

Scope	Configuration	τ	R@1	R@5	R@10	MRR
corpus	baseline	0.5	.026	.075	.111	.047
corpus	baseline	0.3	.054	.130	.186	.089
corpus	+ decompose + rerank	0.5	.030	.087	.128	.055
corpus	+ decompose + rerank	0.3	.062	.151	.208	.101
video	baseline	0.5	.148	.394	.491	.250
video	baseline	0.3	.311	.660	.774	.456
video	+ decompose + rerank	0.5	.154	.410	.501	.258
video	+ decompose + rerank	0.3	.323	.677	.780	.469

Two conclusions follow: the system's practical localisation ability ("lands within a loose overlap of the right moment") is considerably better than the strict headline numbers suggest —within the correct video, a correct moment is in the top five for two-thirds of questions —and the relative ordering of configurations is stable across thresholds, so the ablation conclusions above do not depend on the choice of τ.

5.4 RQ3 —Agentic question answering

5.4.1 Agent comparison

Table 5.4 and Figure 5.3 (fig7_qa_by_type) compare the two agent designs of Chapter 3 on the first 150 grounded val questions under a text-only LLM (deepseek-chat): the single-loop function-calling agent (W4) and the LangGraph state machine (W7), which adds the temporal tool (`get_segments_around`) and a bounded self-reflection pass that audits the draft answer's citations against the gathered evidence before accepting it.

Table 5.4 —Five-choice QA accuracy, 150 val questions (text-only LLM).

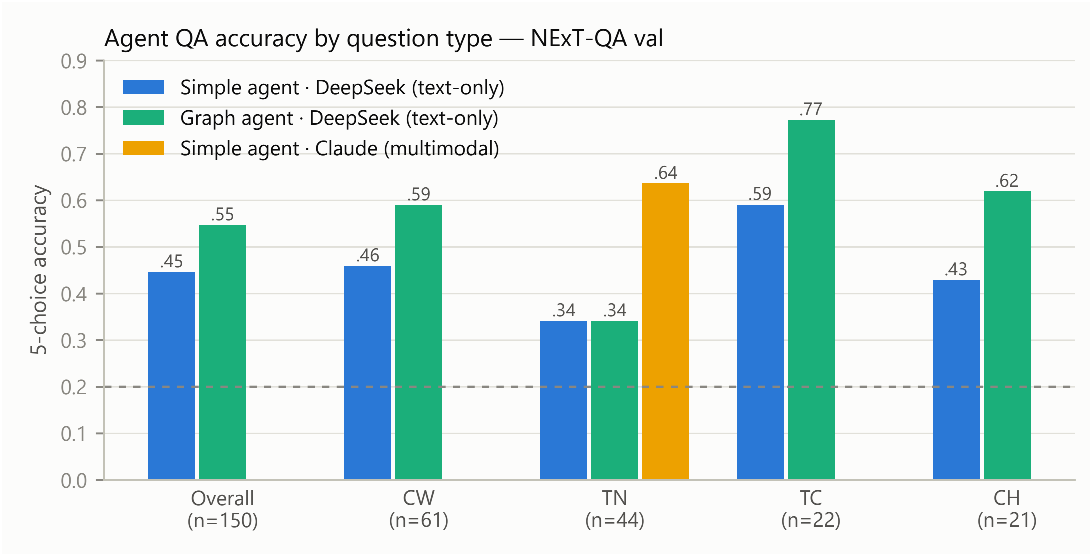

Figure 5.3: Question-answering accuracy by question type and agent configuration.

Agent	Overall	CW (n=61)	TN (n=44)	TC (n=22)	CH (n=21)
Simple loop (W4)	.447	.459	.341	.591	.429
LangGraph (W7)	**.547**	**.590**	.341	**.773**	**.619**

Both agents clear the .20 random floor by a wide margin, confirming that retrieved transcript evidence carries usable signal. The graph agent improves overall accuracy by 10 points (+22% relative), with the gains concentrated in causal questions (CW +13.1, CH +19.0 points) and temporal-current questions (TC +18.2). Inspection of transcripts attributes the gains to three behaviours: systematic anchor-then-walk retrieval on temporal questions, reformulated follow-up searches when initial evidence is thin, and the reflection pass —which in observed runs rejected drafts whose citations did not appear in the gathered evidence, forcing a grounded rewrite.

The striking exception is TN (*what happened next*), frozen at .341 for **both** agents. Qualitative inspection shows the temporal machinery working exactly as designed —the agent anchors the cue event and retrieves the segments that follow it —and then failing at the last step: the answer to a TN question is almost always a *visual action* (what a person or animal does next), the following segments are typically silent, and a text-only LLM receives no usable evidence. In one representative case (“what does the white dog do after going to the cushion”, ground truth *smells the black dog*), the text-only agent responded that the post-anchor segments contain no speech and the action “cannot be determined from the available evidence”—a correct description of its own evidence starvation.

5.4.2 Controlled experiment: visual evidence at the answering stage

The TN result yields a precise hypothesis: the bottleneck is the *evidence channel*, not the temporal mechanism. To test it, the same 44 TN questions were re-run with a single change: the provider was switched to a multimodal LLM (claude-opus-4-8), whose tool results embed each segment’s keyframe image alongside the transcript. Retrieval index, tools, prompts, and questions were held fixed.

Table 5.5 —Same 44 TN questions, evidence channel varied.

Configuration	TN accuracy
Text-only LLM, simple agent	.341
Text-only LLM, graph agent (temporal tool)	.341
Multimodal LLM, keyframes in evidence	**.636**

Accuracy rises by 29.5 points (+87% relative, 44/44 questions completed). On the representative case above, the multimodal agent answered "leans down and sniffs/nuzzles the small black puppy [28−36 s]"—matching the ground truth almost verbatim from the keyframes alone; Figure 5.4 (fig8_whitedog_case) shows the keyframes in question beside the two models' verbatim answers. Because everything except the evidence modality was held constant, the improvement is attributable to visual evidence at the answering stage. Together with Section 5.2, this closes the loop on the modality question at *both* ends of the pipeline: retrieval needs visual embeddings to find the right moment, and generation needs visual evidence to describe it.

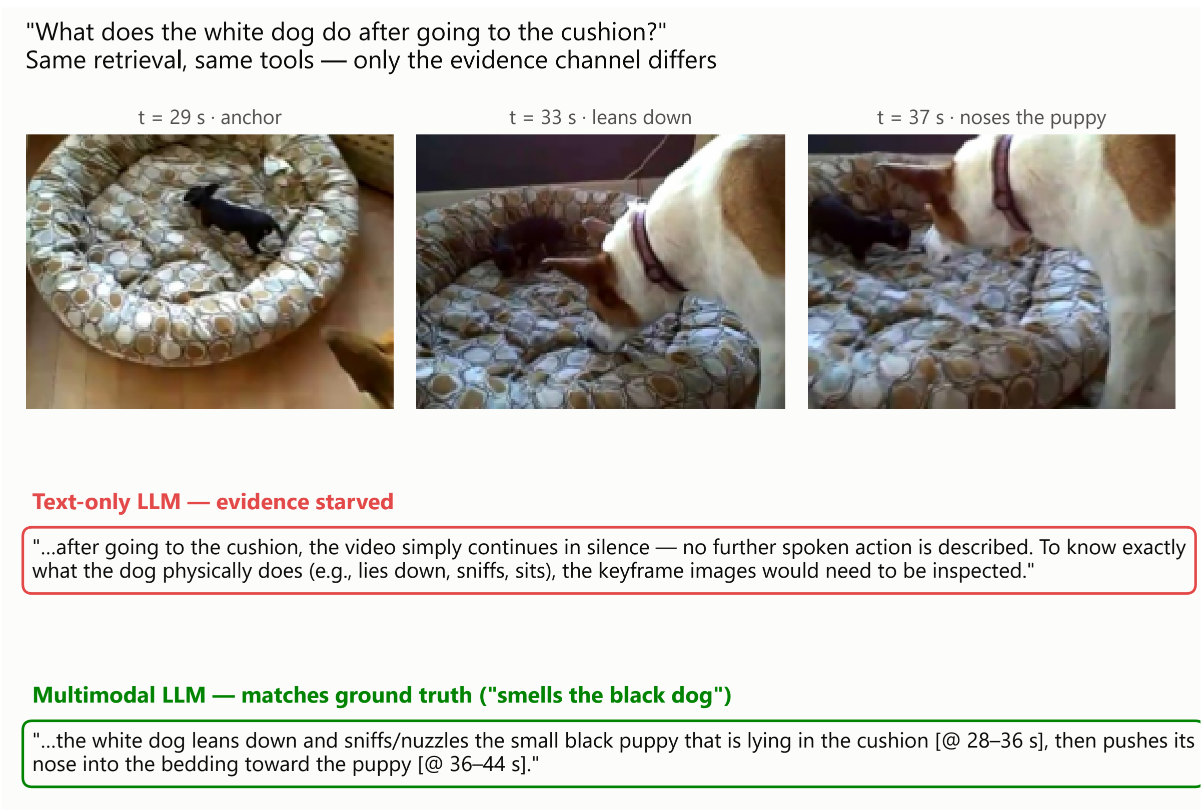

Figure 5.4: White-dog qualitative case: retrieved keyframes and answers under text-only and visual evidence.

5.4.3 Interpreting the .341 floor

Two readings of the text-only TN floor deserve separation. Part of the .14 margin above random reflects genuine transcript evidence; part likely reflects NExT-QA answer priors —a language model can sometimes reject implausible distractors without any evidence. The controlled experiment bounds the effect: whatever fraction of .341 is prior-driven, the additional .295 from keyframes is not, since priors were identical across conditions. A fuller prior-only baseline (answering without any tool access) is left to future work.

5.5 System-level measurements

The offline stage processed all 567 videos in approximately 2 h on a single RTX 4050 laptop GPU (6 GB): keyframe extraction at <1 s per video and Whisper large-v3 (int8, CTranslate2) transcription at 2 − 20 s per video; CLIP ViT-B-32 embedding and indexing of 5,725 segments adds approximately 15 min, and the SigLIP index approximately 35 min. The resulting index is portable (17 MB of vectors per backbone in npz form plus the ChromaDB store) and queries run interactively: question encoding is sub-second on GPU, HNSW search over 5,725 vectors is milliseconds, and the visual re-ranker adds one batched matrix product over 30 candidates. End-to-end agent answers complete in 15 − 60 s depending on tool rounds, dominated by LLM latency. Marginal API costs are small: query decomposition for the entire split cost approximately US$0.15 (cached thereafter), a 150-question text-only QA run approximately US$0.35, and the 44-question multimodal run approximately US$3. The complete evaluation suite of this chapter is reproducible on one consumer laptop plus approximately US$5 of API usage.

5.6 Summary

Against RQ1, multi-modal (visual) retrieval outperforms the text-only baseline by an order of magnitude in corpus scope, and the corpus/video gap identifies cross-video confusion as the dominant error. Against RQ2, the ablations show that (i) naive late fusion cannot rescue a weak text channel at any weight; (ii) query decomposition buys recall at depth while visual re-ranking buys top-rank precision, and they compose; (iii) SigLIP is the strongest index backbone, nearly doubling baseline R@1; and (iv) a two-stage cheap-index/expensive-re-rank design matches the expensive index outright. Against RQ3, a LangGraph agent with temporal tools and self-reflection improves QA accuracy by 10 points over a single-loop agent, and a controlled evidence-channel experiment shows visual evidence nearly doubling temporal-next accuracy —establishing that modality matters at the answering stage independently of retrieval. Chapter 6 discusses the limitations and generality of these findings.

Chapter 6

Chapter 6 —Discussion

6.1 Synthesis: three separable problems

The clearest lesson of Chapter 5 is that the end-to-end failures of a video RAG system decompose into three problems that are *separable* —they have different causes, respond to different interventions, and the interventions compose.

The first is a **recall problem**: the correct moment never enters the candidate pool because the query is phrased in causal or temporal language that a contrastive vision-language encoder cannot match against pixels. Query decomposition addresses exactly this —rewriting the question into scene captions lifted deep-rank recall (corpus R@10 +10% relative) while leaving the top rank untouched (§5.3.2). The second is a **precision problem**: the correct moment is in the pool but near-tied cosine scores cannot order it first. Second-stage re-scoring with a stronger visual model addresses this, and only this (§5.3.3); so does exchanging the index backbone outright (§5.3.4). The third is an **evidence problem**: the correct moment is retrieved, correctly time-stamped, and placed in front of the answerer —which then cannot answer because the decisive evidence is visual and its input is text. No amount of retrieval improvement fixes this; changing the evidence channel does (+87% on temporal-next questions, §5.4.2).

Reading the results this way explains why the interventions compose additively rather than cannibalising each other (Table 5.2): each removes a different bottleneck. It also offers a diagnostic recipe for practitioners: compare corpus against video scope to separate cross-video confusion from localisation error, compare shallow against deep recall to separate recall from precision problems, and compare text-only against multimodal answering to expose evidence starvation.

A second thread worth drawing out is that **the value of a modality is role-dependent, not absolute**. The transcript channel failed twice as a *retrieval key* —as embeddings (§5.2) and as a cross-encoder signal (§5.3.3) —yet the same transcripts, delivered as *evidence* to the answering LLM, sustain QA accuracy far above the random floor (.547 overall) and clearly carry the causal questions, where much of the gain over chance cannot plausibly come from priors alone. Speech in everyday video is a poor pointer to *where* something happens, but once the right moment is found it still describes some of *what* happens. Modality ablations that report a single "contribution" number conflate these two roles; our results suggest they should be measured separately.

Finally, the two-stage equivalence result (§5.3.4) has an architectural implication beyond this project. At 5,725 segments, indexing with the expensive model is trivially affordable, and yet the cheap-index/expensive-re-rank design already matched it to within measurement noise. Since re-ranking cost is fixed (top-30 per query) while indexing cost grows linearly with the corpus, the two-stage design is the only one of the two that survives scaling —and this work provides evidence that the quality sacrifice can be nil.

6.2 Limitations

Single benchmark, single domain. All conclusions are drawn from NExT-QA/GQA: short, everyday, speech-sparse home videos. The negative results about the text channel —fusion never helping (§5.3.1), the text cross-encoder hurting (§5.3.3) —are best read as *conditional on this domain*. In speech-dense, single-language domains (lectures, meetings, news), transcripts are far more informative and the same components may well invert sign. The positive results (decomposition, visual re-ranking, backbone ordering, evidence-channel effect) rest on properties of contrastive encoders and of the QA task rather than of this corpus, and should transfer more readily; verifying this on a second benchmark is the most direct extension of this work.

A confound inside the text-channel result. The weak text channel has two entangled causes: the transcripts are often *uninformative* (chatter unrelated to the questions), and they are often *unreadable* by the predominantly-English CLIP text encoder (multilingual speech). The present experiments cannot apportion blame between the two. A translate-then-embed variant —machine-translating all transcripts to English before indexing —would separate them cheaply and was left undone only for time.

Segmentation granularity bounds temporal accuracy. Segments are fixed 8-second windows on a 4-second stride, while ground-truth intervals vary freely; even a perfect ranker cannot exceed the tIoU that the best-aligned window achieves. Part of the headline gap between τ = 0.5 and τ = 0.3 results (§5.3.5) is this quantisation, not ranking error. tIoU@1 therefore carries a structural ceiling that is a property of the segmenter; adaptive or shot-aligned segmentation would raise the ceiling itself.

Multiple-choice accuracy is a proxy. NExT-QA's five-way format enables cheap, objective scoring, but it measures answer *selection*, not answer *quality*: the faithfulness of the generated prose and the correctness of its timestamp citations are audited only by the agent's own reflection pass, not by human or LLM-as-judge evaluation. The white-dog case study (§5.4.2) illustrates that the generated answers can be precise and well-grounded, but no systematic open-ended-answer evaluation was performed.

Sample sizes and single runs. Retrieval results cover all 3,358 questions and are deterministic given the frozen decomposition cache. The QA experiments, however, are budget-limited: 150 questions for the agent comparison, 44 for the controlled vision experiment, and the text cross-encoder was rejected on a 100-question development subset. All agent numbers are single runs of non-deterministic API models without confidence intervals; the reported effects (+10 points overall, +29.5 points on TN) are large relative to plausible variance, but small differences in Table 5.4's by-type columns ($n \leq 22$) should not be over-read.

Asymmetries in the agent comparison. The LangGraph agent currently targets OpenAI-compatible providers only, so the controlled vision experiment ran under the *simple* agent; the strongest configuration observed (multimodal evidence) and the strongest agent (graph) have never been combined, and the .636 TN result is therefore likely a floor, not a ceiling. Symmetrically, the graph agent's reflection pass and the multimodal channel might interact (e.g. reflection could audit visual claims), which remains untested.

No user study. The project plan scheduled a small user study (W10); it was descoped in favour of the controlled vision experiment and the ablation matrix, which we judged to produce more defensible evidence per unit time. The Streamlit demo (§3.6) stands in for qualitative validation, but claims about end-user utility remain unsupported and are deliberately absent from this dissertation.

6.3 Threats to validity

Construct validity. The hit criterion (correct video and tIoU $\geq \tau$) is stricter than video-level retrieval metrics common in prior work; this deflates absolute numbers relative to that literature but is faithful to the moment-retrieval construct. For QA, the multiple-choice format admits answer-prior effects: a language model can reject implausible distractors without evidence. The controlled experiment bounds this —priors were identical across both evidence conditions, so the +29.5-point effect cannot be prior-driven —but the absolute .341 text-only floor mixes prior with evidence in unknown proportion; a no-retrieval prior-only baseline would complete the picture.

Internal validity. The decomposition cache is shared verbatim across all configurations that use it, eliminating decomposition variance from those comparisons. The re-ranker's fusion weight (0.5) and candidate pool (30) were fixed a priori rather than tuned, and the development subset used to reject the text cross-encoder ($n = 100$) is disjoint in purpose but not in items from the full evaluation set; since the decision it informed was to *exclude* a component, any bias is conservative. Agent experiments called the same LLM family for decomposition and answering (DeepSeek), which could in principle correlate their errors; retrieval experiments are unaffected, as decomposition quality is identical across all retrieval configurations.

External validity. Beyond the domain limitation above, two scale caveats apply. The corpus (567 videos, 5,725 segments) is small enough that exhaustive HNSW search is effectively exact; ANN recall degradation at millions of segments is not measured. And the two-stage equivalence claim is demonstrated at a scale where *both* designs are feasible —its extrapolation to large corpora rests on the argument of §6.1, not on direct measurement. Finally, all API-based results are tied to unversioned commercial model endpoints (deepseek-chat, claude-opus-4-8, mid-2026); the committed prompts, caches, and result files make the analysis auditable, but bit-exact replication of agent runs is not guaranteed by the providers.

6.4 Summary

The findings hold together as a single narrative: in everyday video, visual signal dominates at every stage where a modality choice exists, and the remaining errors are separable into recall, precision, and evidence problems with independent, composable remedies. The principal limitations —one domain, proxy QA metrics, budget-limited agent samples, and an untested graph-plus-vision combination —are all addressable with more compute rather than new machinery, and none threatens the direction of the reported effects, whose magnitudes (+69% relative R@1 from the backbone, +87% from the evidence channel) far exceed the plausible noise floor. Chapter 7 concludes and sets out the corresponding future work.

Chapter 7

Chapter 7 —Conclusion and Future Work

7.1 Answers to the research questions

RQ1 —retrieval effectiveness against text-only baselines. A multi-modal pipeline retrieving on visual embeddings outperforms transcript-based retrieval by an order of magnitude at moment granularity on everyday video (corpus-scope R@10 .111 vs .011 at tIoU ≥ 0.5), and the fully developed system —SigLIP index with LLM query decomposition —nearly doubles the initial visual baseline itself (R@1 .026 → .048; video-scope tIoU@1 .230). Scope analysis attributes the remaining difficulty to cross-video confusion rather than within-video localisation: given the right video, the system places a correct moment in its top five for two-thirds of questions at the looser $\tau = 0.3$ (R@5 .677). The answer to RQ1 is therefore affirmative with a precise caveat: *visual* retrieval is effective and improvable; retrieval on what is *said* is, on this domain, not a viable baseline so much as a cautionary one.

RQ2 —the multi-modal embedding strategy. The experiments support a strategy, not merely a ranking: index the visual channel with the strongest affordable contrastive backbone (SigLIP SO400M dominated both CLIP variants, +69% relative R@1 as the index); if the strongest backbone is unaffordable at corpus scale, index cheaply and re-rank its top candidates with the stronger model, which matched the expensive index to within measurement noise; translate questions into the encoder's native register (scene captions) via LLM decomposition for deep-rank recall; and do *not* fuse in a weak text channel at any weight, nor re-rank with a text cross-encoder —both degrade quality for an identified cause (sparse, multilingual, conversational speech). The optimal strategy for audio and on-screen text on this benchmark is the null strategy; Chapter 6 marks the domains where that conclusion should be re-tested.

RQ3 —agentic decomposition and synthesis. Query decomposition, temporal tool use, and bounded self-reflection each demonstrably contribute. At the retrieval layer, decomposition is safe-by-construction (the original query is always retained) and lifts recall. At the answering layer, a LangGraph state machine that anchors events, walks the timeline, and audits its own citations raises five-way QA accuracy from .447 to .547 over a single-loop agent, with the largest gains on causal and temporal-current questions. The controlled evidence-channel experiment completes the answer: agentic machinery alone cannot compensate for evidence the model cannot perceive

—supplying keyframes to the answerer nearly doubled temporal-next accuracy (.341 → .636) with everything else held fixed. An effective video-QA agent is therefore *jointly* a retrieval planner and a multimodal reader; either half alone leaves the other's headroom unrealised.

7.2 Future work

Six directions follow directly from the limitations identified in Chapter 6, ordered roughly by evidence-per-effort.

Repair the text channel, then re-test fusion. Machine-translate transcripts to English (or adopt a multilingual text-retrieval encoder) before indexing, disentangling the informativeness/readability confound of §6.2 —and only then revisit late fusion and text re-ranking, whose negative results are conditional on the broken channel.

Combine the graph agent with multimodal evidence. The strongest agent and the strongest evidence channel were never run together; extending the LangGraph harness to the Anthropic API (or a vision-capable open model) tests whether reflection can audit *visual* claims, and how far beyond .636 the TN ceiling moves.

Complete the open-source model comparison. The provider abstraction already targets any OpenAI-compatible endpoint; running the QA suite against local Qwen/Llama models (the plumbing and evaluation scripts exist) would quantify the API-versus-open trade-off the project plan anticipated.

A prior-only QA baseline. Answering the multiple-choice questions with no tool access would bound the answer-prior component of all QA numbers (§6.3), sharpening every agent comparison at negligible cost.

Adaptive segmentation. Fixed 8-second windows impose a structural ceiling on tIoU (§6.2); shot-aligned or query-adaptive windowing would raise the ceiling itself, and the evaluation harness can measure the gain directly.

A second domain and a user study. Replicating the modality ablations on speech-dense video (lectures, meetings) tests the conditionality claims of Chapter 6, and the descoped user study remains the right instrument for claims about end-user utility that no offline metric can support.

7.3 Closing remark

The project set out to make a moment of video as findable as a sentence of text, under the discipline that every design choice be measured. Its most useful legacy may be less the headline numbers than the shape of the evidence: failures separated into recall, precision, and evidence problems; negative results retained with their causes; and a pipeline in which each remedy could be switched on, switched off, and priced. Video understanding systems will continue to change quickly; an evaluation harness that can say *which part helped, by how much, and why* will remain the transferable asset.

References

Appendix A

Appendix B —Full per-question-type result tables

Generated by `make_appendix_b.py` from the committed result files in `results/`; regenerate after any re-run. Question types: CW/CH causal, TN/TC/TP temporal. TP has only 52 grounded instances corpus-wide (2 in the 150-question QA sample) —its rows are reported for completeness and should not be interpreted.

A.1 ViT-B-32 baseline —corpus scope (visual, tau 0.5)

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.026	.075	.111	.047	.117	.035
CW (causal why)	1474	.022	.063	.097	.041	.103	.030
TN (temporal next)	879	.025	.073	.107	.045	.107	.032
CH (causal how)	481	.037	.116	.156	.070	.175	.051
TC (temporal current)	472	.030	.078	.121	.052	.124	.041
TP (temporal previous)	52	.000	.019	.096	.013	.070	.015

A.2 ViT-B-32 baseline —video scope (visual, tau 0.5)

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.148	.394	.491	.250	.659	.203
CW (causal why)	1474	.149	.394	.491	.250	.655	.202
TN (temporal next)	879	.129	.379	.495	.230	.621	.177
CH (causal how)	481	.156	.403	.491	.258	.710	.224
TC (temporal current)	472	.182	.424	.494	.283	.697	.235
TP (temporal previous)	52	.096	.308	.442	.187	.585	.151

A.3 Decomposition + visual re-ranking —corpus scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.030	.087	.128	.055	.135	.040
CW (causal why)	1474	.028	.077	.121	.051	.125	.036
TN (temporal next)	879	.031	.078	.107	.051	.119	.034
CH (causal how)	481	.037	.129	.170	.073	.188	.061
TC (temporal current)	472	.030	.097	.144	.061	.143	.044
TP (temporal previous)	52	.019	.058	.115	.035	.110	.031

A.4 Decomposition + visual re-ranking —video scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.154	.410	.501	.258	.670	.209

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
CW (causal why)	1474	.152	.410	.496	.259	.667	.206
TN (temporal next)	879	.133	.396	.513	.238	.631	.183
CH (causal how)	481	.158	.414	.497	.264	.719	.235
TC (temporal current)	472	.195	.439	.508	.293	.704	.243
TP (temporal previous)	52	.135	.346	.442	.212	.627	.184

A.5 SigLIP + decomposition (final best) —corpus scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.048	.110	.161	.076	.167	.055
CW (causal why)	1474	.054	.115	.157	.080	.161	.058
TN (temporal next)	879	.027	.081	.140	.054	.152	.039
CH (causal how)	481	.079	.156	.212	.114	.243	.092
TC (temporal current)	472	.038	.110	.172	.072	.147	.043
TP (temporal previous)	52	.019	.019	.077	.027	.089	.028

A.6 SigLIP + decomposition (final best) —video scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.176	.419	.496	.275	.686	.230

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
CW (causal why)	1474	.193	.427	.495	.289	.695	.243
TN (temporal next)	879	.130	.412	.502	.243	.647	.195
CH (causal how)	481	.189	.410	.486	.280	.718	.247
TC (temporal current)	472	.199	.424	.504	.293	.704	.242
TP (temporal previous)	52	.135	.365	.423	.207	.655	.194

A.7 QA accuracy by type —simple agent, text-only (150 questions)

Type	n	Accuracy
overall	150	.447
CW (causal why)	61	.459
TN (temporal next)	44	.341
CH (causal how)	21	.429
TC (temporal current)	22	.591
TP (temporal previous)	2	1.000

A.8 QA accuracy by type —graph agent, text-only (150 questions)

Type	n	Accuracy
overall	150	.547
CW (causal why)	61	.590
TN (temporal next)	44	.341
CH (causal how)	21	.619
TC (temporal current)	22	.773
TP (temporal previous)	2	.500

A.9 QA accuracy —multimodal simple agent (44 TN questions)

Type	n	Accuracy
overall	44	.636
TN (temporal next)	44	.636